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fi

fi , ([98],[98],[88],..)

2 :

fi. fi. ,(,-, fi ,..)--. 1 .(.,.)-. $_2$fi . 1 .

[91], . ( ) .

ϕ (,) 1, 2( )

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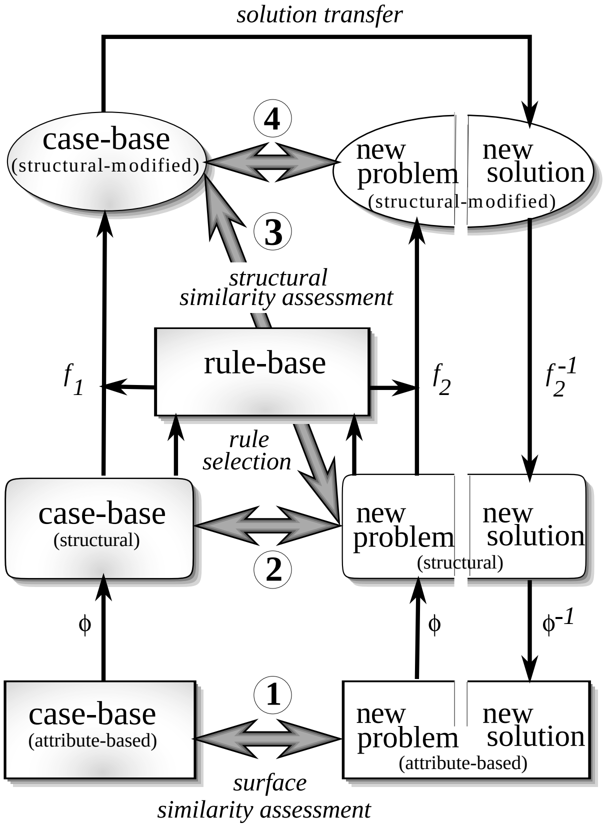
[2] The rules used here are not covered by the expression rewriting rules.


solution transfer
4
case-base
(structural-modified)
new problem
new solution
(structural-modified)
3
structural similarity assessment
rule-base
f_1
f_2
f_2^{-1}
rule selection
case-base
(structural)
2
new problem
new solution
(structural)
φ
φ
ϕ^{-1}
1
case-base
(attribute-based)
new problem
new solution
(attribute-based)
surface similarity assessment

---

[3] In mathematical terms, it is not always possible to invert the mapping ϕ. Thus ϕ^{-1} denotes any computable flattening that takes a structural case description and transforms it into an attribute-based one such that applying ϕ would yield the structured representation, again. The same applies to f and f^{-1}.

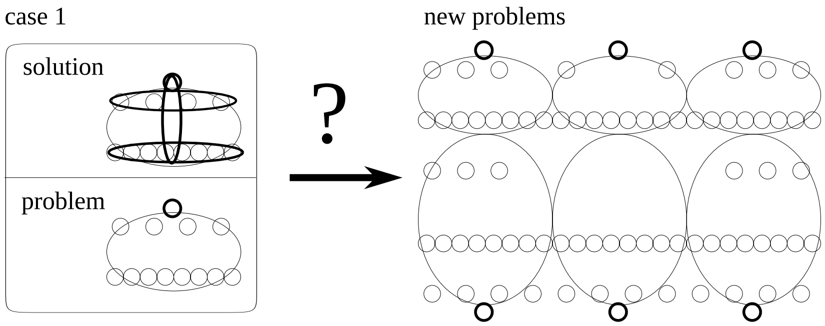
1 ((1)(4)):

(1) -([e1, e2, e2]).

(2)

3 :-

[e3] 2


case 1
solution
problem
?
new problems

2.

3.1

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[4] To get cases in a less redundant form, 'space-coordinates' are normalized. Therefore the smallest x-, y-, and z-dimensions of each case are set to zero.

[..63] . . . ϕ .

3.2 :

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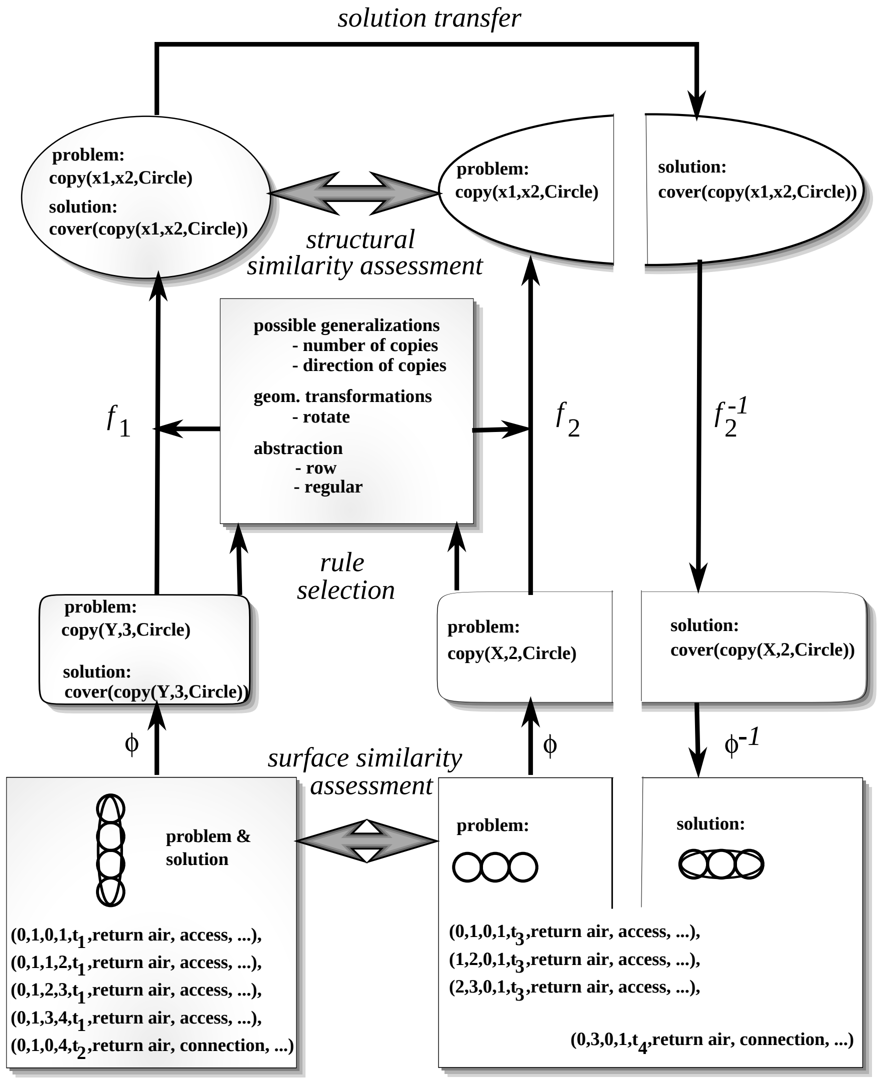
[5] In contrast to, e.g., mechanical engineering where functional dependencies constitute the background knowledge model, in building design topological dependencies play the paramount role and have to be considered during reasoning. Additionally, geometrical laws (reflecting a reflected figure again will result in the original figure) can be exploited.

[6] For simplicity, we only gave the values of the attributes x, dx, y, dy, time, aspect, and morphology of each object.


solution transfer
problem:
copy(x1,x2,Circle)
solution:
cover(copy(x1,x2,Circle))
problem:
copy(x1,x2,Circle)
solution:
cover(copy(x1,x2,Circle))
structural
similarity assessment
possible generalizations
- number of copies
- direction of copies
geom. transformations
- rotate
abstraction
- row
- regular
f_1
f_2
f_2^{-1}
rule
selection
problem:
copy(Y,3,Circle)
solution:
cover(copy(Y,3,Circle))
problem:
copy(X,2,Circle)
solution:
cover(copy(X,2,Circle))
ϕ
ϕ
ϕ^{-1}
surface similarity
assessment
problem &
solution
problem:
solution:
(0,1,0,1,t_1,return air, access, ...),
(0,1,1,2,t_1,return air, access, ...),
(0,1,2,3,t_1,return air, access, ...),
(0,1,3,4,t_1,return air, access, ...),
(0,1,0,4,t_2,return air, connection, ...)
(0,1,0,1,t_3,return air, access, ...),
(1,2,0,1,t_3,return air, access, ...),
(2,3,0,1,t_3,return air, access, ...),
(0,3,0,1,t_4,return air, connection, ...)

-

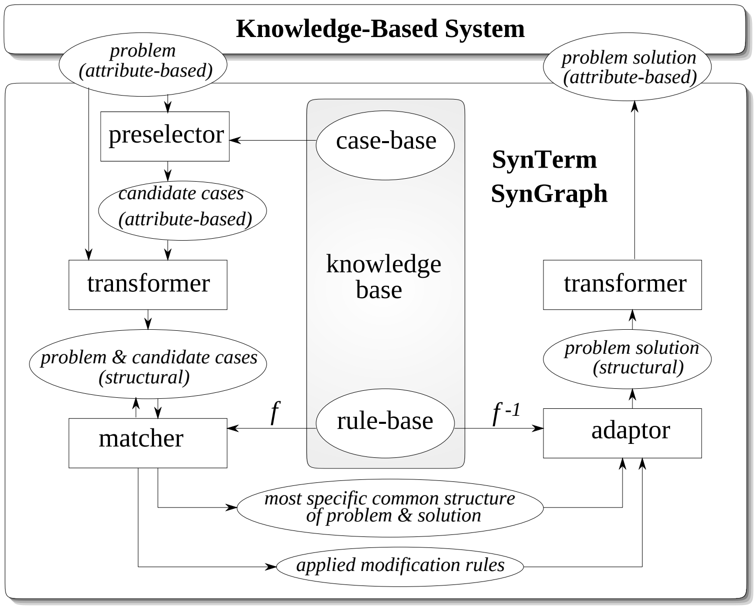
3.3 :

-

ϕ^{-1} - - .

4

(-) (


Knowledge-Based System
problem (attribute-based)
problem solution (attribute-based)
preselector
case-base
SynTerm
SynGraph
candidate cases (attribute-based)
knowledge base
transformer
transformer
problem & candidate cases (structural)
problem solution (structural)
f
rule-base
f^{-1}
matcher
adaptor
most specific common structure of problem & solution
applied modification rules